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Foundations Of Python Programming

Assignment: 05

Course Registration Program Using Dictionary, JSON file and Error Handling

Introduction

This document specified the process of completing an assignment 05 of the Foundation of Python Programming. This assignment is very similar to Assignment 04 where the Python script that allow user to input student's information, display the information and save the data to a JavaScript Object Notation (JSON) file. However, we will be utilizing Python's dictionary to process the data with this assignment/script. Another important item for this assignment is implementing the error handling in the script file. This document will skip the information around how the script is structured and majority of the code are like since the starter script provides you with those implementations. It will mainly focus on the updates we have to do and explain how we complete the assignment with Dictionary, JSON file processing and Error Handling for Python.

High-Level Process

There are total of six (6) steps to complete this assignment:

1.) Add Import Statement and Update Constants and Variables
2.) Read Json File and Implement Error Handle
3.) Update Choice 1: Input User Data Utilizing Dictionary and Implement Error Handle
4.) Update Choice 2: Presenting Data Utilizing Dictionary
5.) Update Choice 3: Save Data to File Utilizing Json File Processing Method and Error Handle
6.) Run and Test Script

Add Import Statement and Update Constants/Variables

In this assignment, the acceptance criteria have specified we first need to add the import the JSON module since the script will be processing a Json file. Line 23 in Figure 1 shows you how to import json module to the script.

The following Constants and Variables also needed to be updated as part of the assignment's acceptance criteria:

- "*FILE_Name: str*" constant is updated from "*Enrollments.csv*" to "*Enrollments.json*"
- "*Student_data: dict*" variable is updated from list to dictional by assigning an empty {}.

The figure 1 shows the updated on a constant and variable that have been mentioned above. The rest of the constant and variables remain unchanged from Assignment 03.

```
import json

# Define the Data Constants
MENU: str = '''
---- Course Registration Program ----
  Select from the following menu:
    1. Register a Student for a Course.
    2. Show current data.
    3. Save data to a file.
    4. Exit the program.
-----------------------------------------
'''
# Define the Data Constants
FILE_NAME: str = "Enrollments.json"

# Define the Data Variables and constants
student_first_name: str = ''  # Holds the first name of a student entered by the user.
student_last_name: str = ''  # Holds the last name of a student entered by the user.
course_name: str = ''  # Holds the name of a course entered by the user.
student_data: dict = {}  # one row of student data
students: list = []  # a table of student data
file = None  # Holds a reference to an opened file.
menu_choice: str  # Hold the choice made by the user.
```

Figure 1: Import Statement and Constant/Variable Update

Read Json File and Implement Error Handle

As part of processing acceptance criteria, the contents of the "*Enrollments.json*" are to be read into "*students: list*" two-dimensional list of dictionary rows using the *json.load()* function. Additionally, the script will need to provides structured error handling when the file is read into the list of dictionary rows. To accomplish the error handling, we will be using try-except statement. This block of the statement will be throwing exceptions when the errors occur and catch them to be display to the console. For reading json file block of code, we will be trying "*FileNotFoundError*" and generic "*Exception*". For this assignment, we will be adding a "finally" statement to handle *file.close()* as well. Figure 2 shows the implementation of reading the json file using *json.load()* function and the try-catch-finally block to handle the system

```python
# When the program starts, read the file data into a list of lists (table)
# Extract the data from the file
try:
    file = open(FILE_NAME, "r")
    students = json.load(file)
except FileNotFoundError as e:
    print(f"File {FILE_NAME} is not found!")
    print("-- Technical Error Message -- ")
    print(e, e.__doc__, type(e), sep='\n')
except Exception as e:
    print("There was a non-specific error!\n")
    print("-- Technical Error Message -- ")
    print(e, e.__doc__, type(e), sep="\n")
finally:
    # Check if a file object exists and is still open
    if file is not None and file.closed == False:
        file.close()

```

Figure 2: Read Json File and Implement Error Handling

Update Choice 1: Input User Data Utilizing Dictionary and Implement Error Handle

As part of this assignment acceptance criteria, menu choice 1 will need to be prompted to get student information just like assignment 03. However, the data collected from the user will be stored in the dictionary with the appropriate key including FirstName, LastName, and CourseName. Then the dictionary with the collected data will be added to a "*students"* two-dimensional list variable of dictionaries rows. Additionally, we will be implementing error handling for when the user enters a fist name and a last name. In this block of code, we will be trying to catch if the user enters non-alphabetic first and last name and if that happens, the script will "*raise*" a "*ValueError"* with an appropriate customize messages. Figure 3 shows the exact implementation that was done to accomplish the menu choice 1 updates.

```python
    # Input user data
    if menu_choice == "1":  # This will not work if it is an integer!
        try:
            student_first_name = input("Enter the student's first name: ")
            if not student_first_name.isalpha():
                raise ValueError("Student first name must be alphanumeric")

            student_last_name = input("Enter the student's last name: ")
            if not student_last_name.isalpha():
                raise ValueError("Student last name must be alphanumeric")

            course_name = input("Please enter the name of the course: ")
            student_data = {"FirstName": student_first_name,
                            "LastName": student_last_name,
                            "CourseName": course_name}
            students.append(student_data)
            print(f"You have registered {student_first_name} {student_last_name} "
                  f"for {course_name}.")

        except ValueError as e:
            print(f"Student's First name and Last name must only be alphanumeric")
            print("-- Technical Error Message -- ")
            print(e, e.__doc__, type(e), sep='\n')
        except Exception as e:
            print("There was a non-specific error!\n")
            print("-- Technical Error Message -- ")
            print(e, e.__doc__, type(e), sep="\n")
        continue

```

Figure 3: Menu Choice 1 Data Collection Updates and Error Handling Implementation

Update Choice 2: Presenting Data Utilizing Dictionary

As part of "Input/Output" acceptance criteria of assignment 05, the menu choice 2 will need to be updated to present a string by formatting the collected data which is now a student information in row of dictionary inside "*students"* two-dimensional list variable using the "*print()*" function. To meet these criteria, the following changes show in Figure 4 will need to be carried out.

```python
    # Present the current data
    elif menu_choice == "2":

        # Process the data to create and display a custom message
        print("-"*50)
        for student in students:
            print(f"Student {student["FirstName"]} {student["LastName"]} "
                  f"is enrolled in {student["CourseName"]}")
        print("-"*50)
        continue

```

Figure 4: Menu Choice 2 Presenting Data Updates

Update Choice 3: Save Data to File Utilizing Json File Processing Method and Error Handle

For the choice 3, the script will opens a file named "*Enrollments.json"* which has been assigned to "*FILE_NAME"* constant in write mode using the "*open()"* function. This will write the collected contents of the "*students"* list variable to the file using the "*json.dump()"* function. Similar to the updates in the above sections of the script, we will need to implement error handling when the dictionary rows are written to the file. The try-block will throw "*TypeError"* exception if the data intents for json file is invalid. Additionally, the try-block will throw the generical exception for other error that might occur to this block of code. Figure 5 shows the write to json file implementation and error handling implementation.

```python
    # Save the data to a file
    elif menu_choice == "3":
        try:
            file = open(FILE_NAME, "w")
            json.dump(students, file, indent=2)
        except TypeError as e:
            print("please check that the data is a valid json file!\n")
            print("-- Technical Error Message -- ")
            print(e, e.__doc__, type(e), sep="\n")
        except Exception as e:
            print("-- Technical Error Message -- ")
            print("Built-In Python error info: ")
            print(e, e.__doc__, type(e), sep="\n")
            # Check if a file object exists and is still open
        finally:
            if file is not None and file.closed == False:
                file.close()

        print("The following data was saved to file!")
        for student in students:
            print(f"Student {student["FirstName"]} {student["LastName"]} "
                  f"is enrolled in {student["CourseName"]}")
        continue
```

Figure 5: Menu Choice 2 Saving Data to Json File Updates

Run and Test Script

This section will include the script run to test whether the implementation produces the result as we expected. The acceptance criteria have provided seven (7) total test cases for us to run. Figure 6 shows us the results for single user input and Figure 8 show us the results for multiple user inputs while running on PyCharm. Figure 7 and Figure 9 shows the "*Enrollments.json*" file after writing additional data to it on PyCharm. Figure 10 shows us the results for single user input and Figure 12 show us the results for multiple user inputs while running on Windows Console. Figure 11 and Figure 13 shows the "*Enrollments.json*" file after writing additional data to it on Windows Console.

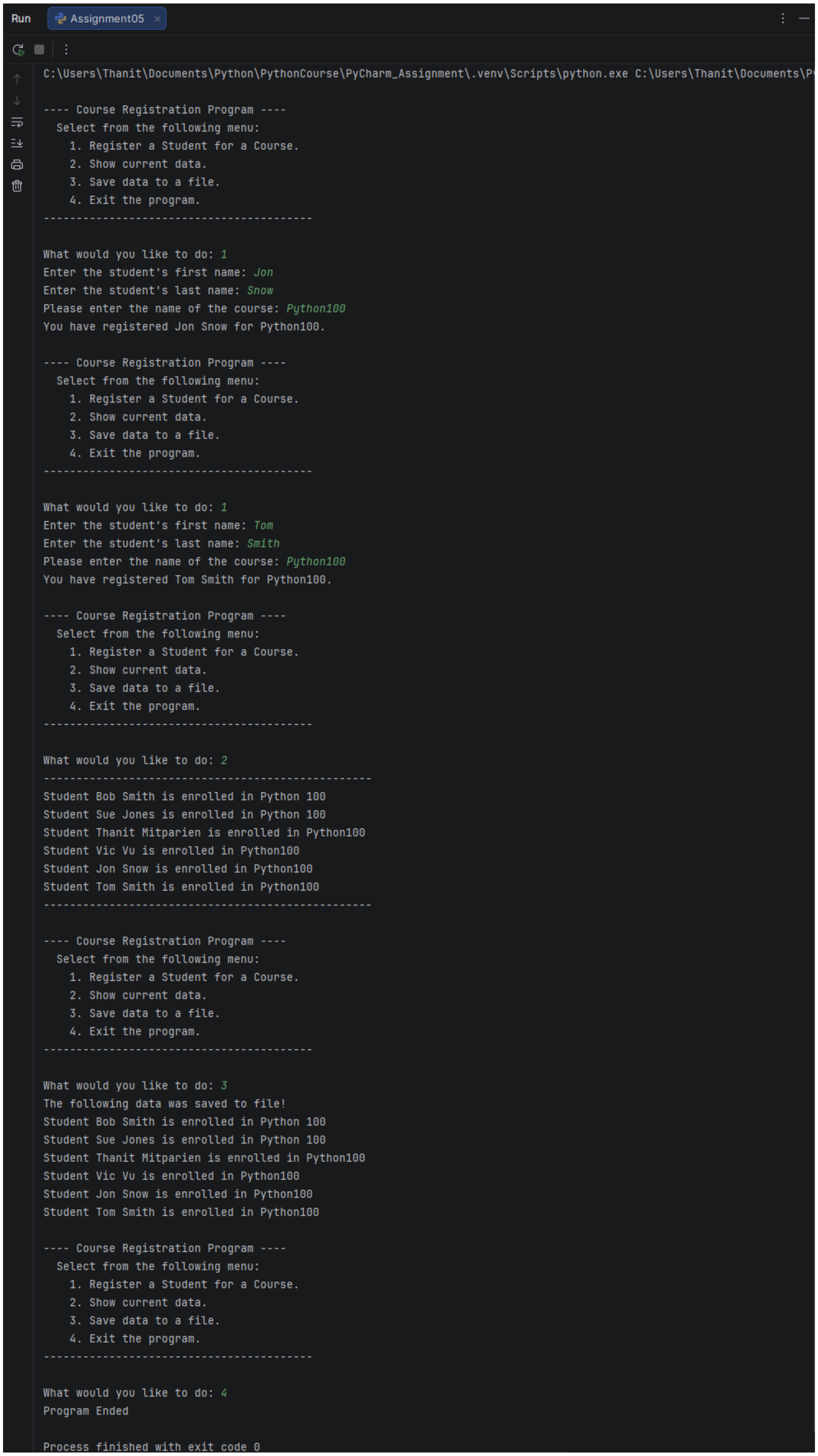
```
Run    Assignment05

C:\Users\Thanit\Documents\Python\PythonCourse\PyCharm_Assignment\.venv\Scripts\python.exe C:\Users\Thanit\Documents\Py

---- Course Registration Program ----
  Select from the following menu:
    1. Register a Student for a Course.
    2. Show current data.
    3. Save data to a file.
    4. Exit the program.
-----------------------------------------

What would you like to do: 1
Enter the student's first name: Thanit
Enter the student's last name: Mitparien
Please enter the name of the course: Python100
You have registered Thanit Mitparien for Python100.

---- Course Registration Program ----
  Select from the following menu:
    1. Register a Student for a Course.
    2. Show current data.
    3. Save data to a file.
    4. Exit the program.
-----------------------------------------

What would you like to do: 2
--------------------------------------------------
Student Bob Smith is enrolled in Python 100
Student Sue Jones is enrolled in Python 100
Student Thanit Mitparien is enrolled in Python100
--------------------------------------------------

---- Course Registration Program ----
  Select from the following menu:
    1. Register a Student for a Course.
    2. Show current data.
    3. Save data to a file.
    4. Exit the program.
-----------------------------------------

What would you like to do: 3
The following data was saved to file!
Student Bob Smith is enrolled in Python 100
Student Sue Jones is enrolled in Python 100
Student Thanit Mitparien is enrolled in Python100

---- Course Registration Program ----
  Select from the following menu:
    1. Register a Student for a Course.
    2. Show current data.
    3. Save data to a file.
    4. Exit the program.
-----------------------------------------

What would you like to do: 4
Program Ended

Process finished with exit code 0
```

Figure 6: Single User's input Run/Test on PyCharm

Assignment05.py   Enrollments.json

```json
[
  {
    "FirstName": "Bob",
    "LastName": "Smith",
    "CourseName": "Python 100"
  },
  {
    "FirstName": "Sue",
    "LastName": "Jones",
    "CourseName": "Python 100"
  },
  {
    "FirstName": "Thanit",
    "LastName": "Mitparien",
    "CourseName": "Python100"
  }
]
```

Figure 7: Single User's Input Enrollments.json File Update on PyCharm Run



```
Run    Assignment05

C:\Users\Thanit\Documents\Python\PythonCourse\PyCharm_Assignment\.venv\Scripts\python.exe C:\Users\Thanit\Documents\P

---- Course Registration Program ----
  Select from the following menu:
    1. Register a Student for a Course.
    2. Show current data.
    3. Save data to a file.
    4. Exit the program.
-----------------------------------------

What would you like to do: 1
Enter the student's first name: Jon
Enter the student's last name: Snow
Please enter the name of the course: Python100
You have registered Jon Snow for Python100.

---- Course Registration Program ----
  Select from the following menu:
    1. Register a Student for a Course.
    2. Show current data.
    3. Save data to a file.
    4. Exit the program.
-----------------------------------------

What would you like to do: 1
Enter the student's first name: Tom
Enter the student's last name: Smith
Please enter the name of the course: Python100
You have registered Tom Smith for Python100.

---- Course Registration Program ----
  Select from the following menu:
    1. Register a Student for a Course.
    2. Show current data.
    3. Save data to a file.
    4. Exit the program.
-----------------------------------------

What would you like to do: 2
--------------------------------------------------
Student Bob Smith is enrolled in Python 100
Student Sue Jones is enrolled in Python 100
Student Thanit Mitparien is enrolled in Python100
Student Vic Vu is enrolled in Python100
Student Jon Snow is enrolled in Python100
Student Tom Smith is enrolled in Python100
--------------------------------------------------

---- Course Registration Program ----
  Select from the following menu:
    1. Register a Student for a Course.
    2. Show current data.
    3. Save data to a file.
    4. Exit the program.
-----------------------------------------

What would you like to do: 3
The following data was saved to file!
Student Bob Smith is enrolled in Python 100
Student Sue Jones is enrolled in Python 100
Student Thanit Mitparien is enrolled in Python100
Student Vic Vu is enrolled in Python100
Student Jon Snow is enrolled in Python100
Student Tom Smith is enrolled in Python100

---- Course Registration Program ----
  Select from the following menu:
    1. Register a Student for a Course.
    2. Show current data.
    3. Save data to a file.
    4. Exit the program.
-----------------------------------------

What would you like to do: 4
Program Ended

Process finished with exit code 0
```

Figure 8: Multiple User's inputs Run/Test on PyCharm

```json
[
  {
    "FirstName": "Bob",
    "LastName": "Smith",
    "CourseName": "Python 100"
  },
  {
    "FirstName": "Sue",
    "LastName": "Jones",
    "CourseName": "Python 100"
  },
  {
    "FirstName": "Thanit",
    "LastName": "Mitparien",
    "CourseName": "Python100"
  },
  {
    "FirstName": "Vic",
    "LastName": "Vu",
    "CourseName": "Python100"
  },
  {
    "FirstName": "Jon",
    "LastName": "Snow",
    "CourseName": "Python100"
  },
  {
    "FirstName": "Tom",
    "LastName": "Smith",
    "CourseName": "Python100"
  }
]
```

Figure 9: Multiple User's Input Enrollments.json File Update on PyCharm Run

```
Command Prompt

C:\Users\Thanit\Documents\Python\PythonCourse\PyCharm_Assignment>python Assignment05.py

---- Course Registration Program ----
  Select from the following menu:
    1. Register a Student for a Course.
    2. Show current data.
    3. Save data to a file.
    4. Exit the program.
-----------------------------------------

What would you like to do: 1
Enter the student's first name: Vic
Enter the student's last name: Vu
Please enter the name of the course: Python100
You have registered Vic Vu for Python100.

---- Course Registration Program ----
  Select from the following menu:
    1. Register a Student for a Course.
    2. Show current data.
    3. Save data to a file.
    4. Exit the program.
-----------------------------------------

What would you like to do: 2
--------------------------------------------------
Student Bob Smith is enrolled in Python 100
Student Sue Jones is enrolled in Python 100
Student Thanit Mitparien is enrolled in Python100
Student Vic Vu is enrolled in Python100
--------------------------------------------------

---- Course Registration Program ----
  Select from the following menu:
    1. Register a Student for a Course.
    2. Show current data.
    3. Save data to a file.
    4. Exit the program.
-----------------------------------------

What would you like to do: 3
The following data was saved to file!
Student Bob Smith is enrolled in Python 100
Student Sue Jones is enrolled in Python 100
Student Thanit Mitparien is enrolled in Python100
Student Vic Vu is enrolled in Python100

---- Course Registration Program ----
  Select from the following menu:
    1. Register a Student for a Course.
    2. Show current data.
    3. Save data to a file.
    4. Exit the program.
-----------------------------------------

What would you like to do: 4
Program Ended

C:\Users\Thanit\Documents\Python\PythonCourse\PyCharm_Assignment>
```

Figure 10: Single User's input Run/Test on Windows Console

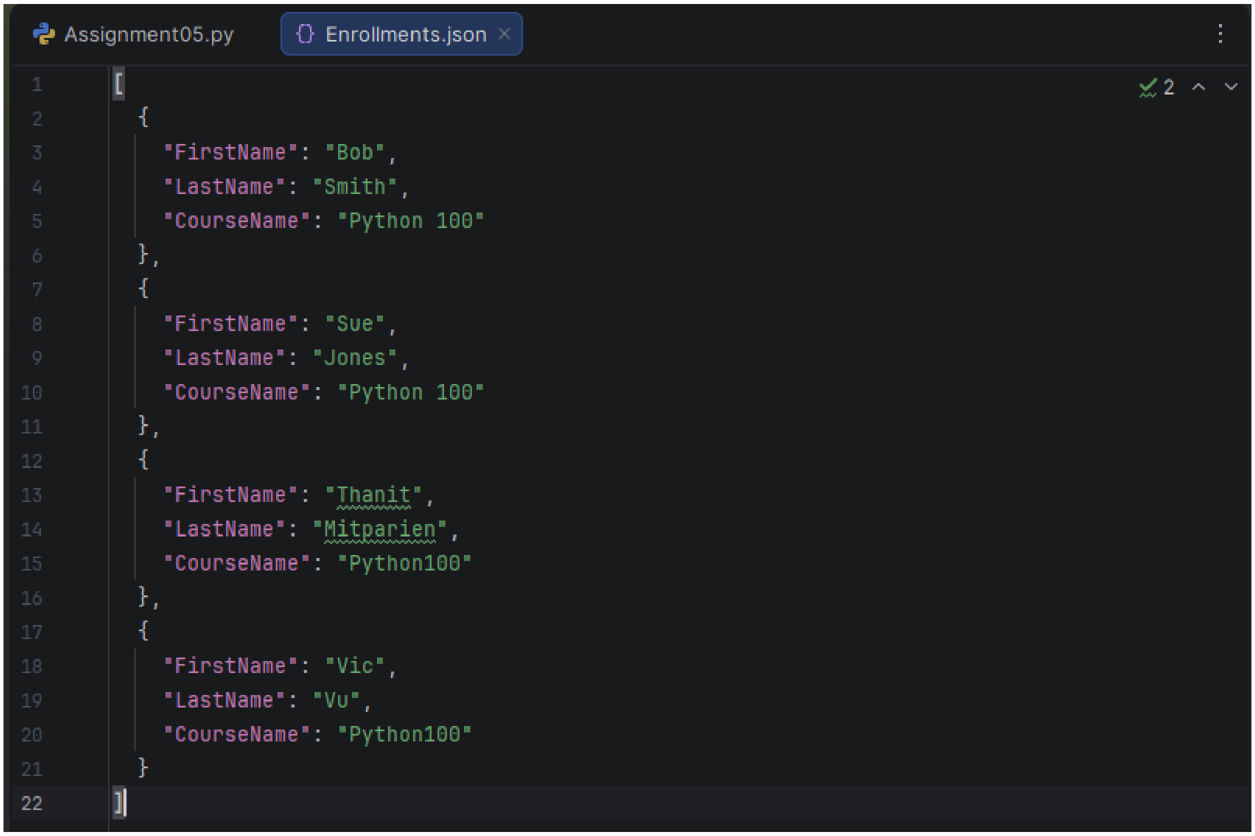


Figure 11: Single User's Input Enrollments.json File Update on Windows Console Run

```
Command Prompt

C:\Users\Thanit\Documents\Python\PythonCourse\PyCharm_Assignment>Python Assignment05.py

---- Course Registration Program ----
  Select from the following menu:
    1. Register a Student for a Course.
    2. Show current data.
    3. Save data to a file.
    4. Exit the program.
-----------------------------------------

What would you like to do: 1
Enter the student's first name: Victor
Enter the student's last name: Choi
Please enter the name of the course: Python100
You have registered Victor Choi for Python100.

---- Course Registration Program ----
  Select from the following menu:
    1. Register a Student for a Course.
    2. Show current data.
    3. Save data to a file.
    4. Exit the program.
-----------------------------------------

What would you like to do: 1
Enter the student's first name: Edward
Enter the student's last name: Snow
Please enter the name of the course: Python100
You have registered Edward Snow for Python100.

---- Course Registration Program ----
  Select from the following menu:
    1. Register a Student for a Course.
    2. Show current data.
    3. Save data to a file.
    4. Exit the program.
-----------------------------------------

What would you like to do: 2
--------------------------------------------------
Student Bob Smith is enrolled in Python 100
Student Sue Jones is enrolled in Python 100
Student Thanit Mitparien is enrolled in Python100
Student Vic Vu is enrolled in Python100
Student Jon Snow is enrolled in Python100
Student Tom Smith is enrolled in Python100
Student Victor Choi is enrolled in Python100
Student Edward Snow is enrolled in Python100
--------------------------------------------------

---- Course Registration Program ----
  Select from the following menu:
    1. Register a Student for a Course.
    2. Show current data.
    3. Save data to a file.
    4. Exit the program.
-----------------------------------------

What would you like to do: 3
The following data was saved to file!
Student Bob Smith is enrolled in Python 100
Student Sue Jones is enrolled in Python 100
Student Thanit Mitparien is enrolled in Python100
Student Vic Vu is enrolled in Python100
Student Jon Snow is enrolled in Python100
Student Tom Smith is enrolled in Python100
Student Victor Choi is enrolled in Python100
Student Edward Snow is enrolled in Python100

---- Course Registration Program ----
  Select from the following menu:
    1. Register a Student for a Course.
    2. Show current data.
    3. Save data to a file.
    4. Exit the program.
-----------------------------------------

What would you like to do: 4
Program Ended

C:\Users\Thanit\Documents\Python\PythonCourse\PyCharm_Assignment>
```

Figure 12: Multiple User's inputs Run/Test on Windows Console

Assignment05.py   Enrollments.json


```json
[
  {
    "FirstName": "Bob",
    "LastName": "Smith",
    "CourseName": "Python 100"
  },
  {
    "FirstName": "Sue",
    "LastName": "Jones",
    "CourseName": "Python 100"
  },
  {
    "FirstName": "Thanit",
    "LastName": "Mitparien",
    "CourseName": "Python100"
  },
  {
    "FirstName": "Vic",
    "LastName": "Vu",
    "CourseName": "Python100"
  },
  {
    "FirstName": "Jon",
    "LastName": "Snow",
    "CourseName": "Python100"
  },
  {
    "FirstName": "Tom",
    "LastName": "Smith",
    "CourseName": "Python100"
  },
  {
    "FirstName": "Victor",
    "LastName": "Choi",
    "CourseName": "Python100"
  },
  {
    "FirstName": "Edward",
    "LastName": "Snow",
    "CourseName": "Python100"
  }
]
```


Figure 13: Multiple User's Input Enrollments.json File Update on Windows Console Run

Summary

This report documents the completion of Assignment 05 for the Foundation of Python Programming course. The assignment builds upon the foundation established in Assignment 03 and Assignment 04, extending the previous student information management script by utilizing dictionary to process and manage data more efficiently along with method to process JSON file. Additionally, this assignment also contains the error handling method.

The core functionality of the script remains consistent with its predecessor, enabling users to input student information, display records, and export data to a Comma-Separated Values (CSV) file. The key distinction of this assignment lies in the application of dictionary and JSON file handling as the primary mechanisms for data processing, replacing earlier, more basic implementations. The error handling is also one of the skills we would need as a programmer to deal with errors and bugs that might occur in our code.

As the starter script provides the foundational structure and majority of the base code, this document does not revisit those elements in detail. Instead, it focuses specifically on the modifications and additions made to fulfill the assignment requirements, offering clear explanations of how lists and loops were implemented to achieve the desired functionality.